

Backtesting Probability of Default Models

The Multi-Period Average Testing in the Presence of Serial Correlation
in Observed Default Rates

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Internal Ratings-Based Modeling Process

- The development of an Internal Ratings-Based (IRB) model consists of two main areas: the development of the risk differentiation function and risk quantification.
- Development of the risk differentiation function is the first step in the overall model development process. The main goal of this step is to ensure the model can adequately rank risk, providing meaningful differentiation between low- and high-risk obligors or facilities.
- After developing the risk differentiation function, practitioners perform the risk quantification process, which essentially involves assigning an appropriate risk level to the output of the risk differentiation function.
- Risk quantification typically consists of specific steps such as: assignment of the base or best estimate, quantification of appropriate adjustments (AA), quantification of the margin of conservatism (MoC), and quantification of any other additional conservatism, for instance that required by certain local regulators. The first two steps are typically referred to as calibration.
- These steps apply to all three risk parameters: Probability of Default (PD), Loss Given Default (LGD), and Exposure at Default (EAD).

Probability of Default Backtesting

- In practice, backtesting is one of the most comprehensive validation exercises, typically investigating model estimates across different portfolio subsegments and time dimensions.
- Backtesting refers to assessing the appropriateness of model estimates, with the primary purpose of ensuring that they accurately predict the occurrence of defaults, i.e., that PD estimates provide reliable forecasts of default rates.
- One commonly applied approach is to backtest estimates over a recent period of three to five years, employing the z-score test.
- The z-score test is traditionally used in both one-sided and two-sided versions, often to assess evidence of underestimation or the accuracy (sometimes referred to as precision) of the PD estimate compared to the observed default rate.
- As necessary inputs, the z-score test requires the observed default rate, the calibrated or tested PD, and the sample size or number of exposures used for the test.
- Given that the calibrated PD is typically defined as the arithmetic average of yearly default rates, the question arises of how to properly construct multi-period inputs (the observed default rate and sample size) to align with the method used to calibrate the PD.

Multi-Period Estimator

- Running the z-score test for the purpose of PD backtesting requires the observed default rate, the calibrated or tested PD, and the sample size or number of observations. However, since the calibrated PD is provided by the model and calculated as an arithmetic average of yearly default rates, the other inputs need to align with this approach to ensure a statistically valid hypothesis testing framework.
- Andrija Djurovic proposes a method that incorporates the specificities of the calibrated PD.
- The multi-period estimator has the general form $\frac{1}{T} \sum_{i=1}^T ODR_i$, where ODR_i is the observed default rate at reference date i , and T is the number of reference dates.
- It can be shown that the variance of this estimator is $\frac{\overline{ODR} \cdot (1 - \overline{ODR})}{T^2} \cdot \sum_{i=1}^T \frac{1}{n_i}$.
- Under the assumption that the average observed default rate follows a binomial proportion, its variance is given by $\frac{\overline{ODR}(1 - \overline{ODR})}{N_e}$, where N_e is the effective sample size.

Multi-Period Estimator cont.

- By equating the variance of the average default rate with the variance of the estimator, practitioners obtain the effective sample size as $\frac{T^2}{\sum_{i=1}^T \frac{1}{n_i}}$.
- Practitioners should note that this formula for the effective sample size assumes default rates are independent over time, while defaults within each period follow a binomial distribution. This approach can be extended to incorporate different correlation structures for default rates over time.
- By adjusting the estimator variance for correlation in observed default rates over time through covariance term(s), and equating it to the variance of the average default rate, it can be shown that the effective sample size becomes $\frac{T^2}{\sum_{t=1}^T \frac{1}{n_t} + 2 \sum_{i < j} \frac{\rho_{|i-j|}}{\sqrt{n_i n_j}}}$, where $\rho_{|i-j|}$ denotes the autocorrelation function at lag $|i - j|$.
- As can be seen from the formula, the correlation of default rates over time affects the effective sample size and, consequently, the statistical testing procedure of the z-score test used for multi-period average testing.
- To demonstrate the impact of correlation in default rates over time on the statistical properties of the multi-period testing method, this presentation shows results from a simulation study comparing the distribution of p-values under the true null hypothesis for an approach that incorrectly assumes independence of default rates and one that properly accounts for the correlation structure. In essence, these two approaches differ in how the effective sample size is calculated, as shown in the formulas on this slide.

Simulation Study

The purpose of this simulation is to compare the distribution of p-values under the true null hypothesis for multi-period average testing using two methods. The main difference between these two methods relates to the assumption of independence or dependence of observed default rates over time. Accordingly, the effective sample size for the statistical test is calculated.

The simulation is based on the following inputs:

- 1 The number of exposures per reference date: 143, 122, 212, 190, and 361.
- 2 The average default rate is set equal to the calibrated PD at 5%.
- 3 The observed default rates follows a first-order autocorrelation (ρ) process with an autocorrelation coefficient of 0.50.

The tested methods are labeled as follows:

- 1 *Method 1 (Correlated ODRs)*: this method accounts for the dependence structure of observed default rates when calculating the effective sample size. Specifically, a first-order autocorrelation structure is assumed, as specified in the simulation setup.
- 2 *Method 2 (Independent ODRs)*: this method incorrectly assumes independence between observed default rates and accordingly calculates the effective sample size.

Simulation Study cont.

Given the inputs and the description of the tested methods from the previous slides, the following steps outline the main simulation setup:

- 1 Select the number of reference dates (T) for testing. In this case, it ranges from 2 to 5, with the number of exposures determined by the order shown on the previous slide. For example, if $T = 3$, then the number of exposures per reference date is: 143, 122, 212. The same logic applies to any other T in the range from 2 to 5.
- 2 For the selected T, given the number of exposures and the autocorrelation coefficient, calculate the effective sample size under the assumption of independence and dependence of the observed default rates.
- 3 For each reference date, given the average PD, first-order correlation coefficient (ρ), and number of exposures per reference date (n_i), simulate the default rates from a multivariate normal distribution with mean equal to PD and variance-covariance matrix given as $\Sigma_{ij} = \rho^{|i-j|} \sqrt{\frac{PD(1-PD)}{n_i} \cdot \frac{PD(1-PD)}{n_j}}$.
- 4 Based on the simulated defaults from step 3, calculate the simulated average observed default rate.
- 5 Given the effective sample sizes from step 2, the simulated average default rate from step 3, and the selected true PD, run a z-score test for both methods and different sample sizes.
- 6 Repeat steps 3 to 5 a total of 100,000 times and record the p-values.
- 7 Calculate the Type I error as the proportion of simulated p-values below the 5% significance level.

Given the above setup, for the method that correctly accounts for dependence of default rates over time, the expectation is that the simulated p-values are approximately uniformly distributed and that the Type I error is approximately equal to the selected significance level of 5%.

The following slides present the basics of the z-score test used in this simulation, as well as the testing results.

Simulation Study cont.

The Z-Score Test

The test statistic of the z-score test is given as follows:

$$z_{score} = \frac{ODR - PD}{\sqrt{\frac{PD(1-PD)}{N}}}$$

where:

- ODR is the observed default rate;
- PD is the calibrated PD;
- N is the number of exposures.

Under the assumption that the z_{score} test statistic follows a standard normal distribution, a p-value is calculated accordingly.

Simulation Results

##	T	pd	method	type_I_error
##	2	5.00%	Correlated ODRs	5.1090%
##	2	5.00%	Independent ODRs	11.0330%
##	3	5.00%	Correlated ODRs	4.9170%
##	3	5.00%	Independent ODRs	14.7230%
##	4	5.00%	Correlated ODRs	4.9660%
##	4	5.00%	Independent ODRs	17.0860%
##	5	5.00%	Correlated ODRs	5.0720%
##	5	5.00%	Independent ODRs	18.9880%

Simulation Results cont.

